Test Series No.

1.8

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 03/03/2026

ANSWER KEY

1.	(1)	46.	(2)	91.	(2)	136.	(3)
2.	(1)	47.	(3)	92.	(2)	137.	(2)
3.	(2)	48.	(3)	93.	(3)	138.	(1)
4.	(3)	49.	(1)	94.	(3)	139.	(1)
5.	(3)	50.	(1)	95.	(2)	140.	(2)
6.	(2)	51.	(1)	96.	(4)	141.	(4)
7.	(1)	52.	(1)	97.	(2)	142.	(2)
8.	(3)	53.	(3)	98.	(3)	143.	(1)
9.	(2)	54.	(2)	99.	(1)	144.	(1)
10.	(2)	55.	(2)	100.	(2)	145.	(1)
11.	(4)	56.	(2)	101.	(2)	146.	(4)
12.	(2)	57.	(2)	102.	(2)	147.	(1)
13.	(2)	58.	(3)	103.	(2)	148.	(4)
14.	(1)	59.	(4)	104.	(2)	149.	(1)
15.	(4)	60.	(3)	105.	(3)	150.	(3)
16.	(4)	61.	(2)	106.	(3)	151.	(4)
17.	(2)	62.	(3)	107.	(1)	152.	(1)
18.	(2)	63.	(1)	108.	(1)	153.	(4)
19.	(2)	64.	(4)	109.	(3)	154.	(4)
20.	(1)	65.	(2)	110.	(3)	155.	(2)
21.	(4)	66.	(3)	111.	(4)	156.	(1)
22.	(3)	67.	(3)	112.	(2)	157.	(2)
23.	(1)	68.	(1)	113.	(3)	158.	(3)
24.	(2)	69.	(3)	114.	(2)	159.	(4)
25.	(3)	70.	(2)	115.	(2)	160.	(1)
26.	(3)	71.	(4)	116.	(1)	161.	(3)
27.	(2)	72.	(1)	117.	(4)	162.	(4)
28.	(2)	73.	(1)	118.	(2)	163.	(4)
29.	(4)	74.	(4)	119.	(4)	164.	(2)
30.	(3)	75.	(1)	120.	(3)	165.	(4)
31.	(2)	76.	(1)	121.	(2)	166.	(2)
32.	(1)	77.	(2)	122.	(4)	167.	(3)
33.	(2)	78.	(2)	123.	(2)	168.	(4)
34.	(2)	79.	(3)	124.	(3)	169.	(2)
35.	(2)	80.	(4)	125.	(1)	170.	(1)
36.	(1)	81.	(1)	126.	(4)	171.	(2)
37.	(4)	82.	(2)	127.	(2)	172.	(3)
38.	(4)	83.	(3)	128.	(2)	173.	(3)
39.	(3)	84.	(3)	129.	(2)	174.	(4)
40.	(3)	85.	(2)	130.	(4)	175.	(4)
41.	(1)	86.	(4)	131.	(4)	176.	(4)
42.	(3)	87.	(4)	132.	(4)	177.	(1)
43.	(1)	88.	(4)	133.	(3)	178.	(2)
44.	(2)	89.	(4)	134.	(2)	179.	(3)
45.	(3)	90.	(3)	135.	(3)	180.	(3)

[Contd...

Hints & Solutions

1. (1)

Potential Difference: The voltage across the parallel branches is determined by the bottom branch: $V = I \times R = 2\text{A} \times 3\Omega = 6\text{V}$

Internal Resistance (r): Total current in the top branch is $i = 1\text{A} + 2\text{A} = 3\text{A}$. Using the terminal voltage equation ($V = \mathcal{E} - ir$):

$12 - 3r = 6 \Rightarrow 3r = 6 \Rightarrow r = 2\Omega$

EMF (E): For the middle branch, the potential difference is also 6V . Moving from left to right (against the 1A current): $+1(1) + E = 6$

$\Rightarrow E = 5\text{V}$

(NCERT Ed. 2025-26, 12th, Page No. 97)

2. (1)

Water and biological tissues are diamagnetic. Diamagnetic materials are weakly repelled by magnetic fields. In a sufficiently strong magnetic field, this upward repulsive force can balance gravity, allowing the frog to levitate.

(NCERT Ed. 2025-26, 12th, Page No. 147)

3. (2)

For a system of parallel slabs, we can apply Snell's Law directly between the first and the last medium, as the intermediate medium does not affect the final relation between the angles.

Using Snell's Law: $\mu_1 \sin i = \mu_3 \sin r$

Rearranging the terms to find the ratio: $\frac{\sin i}{\sin r} = \frac{\mu_3}{\mu_1}$

(NCERT Ed. 2025-26, 12th, Page No. 229)

4. (3)

For a linear variation starting from zero ($i = kt$), the mean value is half the maximum value.

$$\text{Average} = \frac{\text{Area under curve}}{\text{Time}} = \frac{\frac{1}{2} \cdot t \cdot I_0}{t} = \frac{I_0}{2}$$

(NCERT Ed. 2025-26, 12th, Page No. 178)

5. (3)

Pole strength (m): Remains unchanged when a magnet is cut perpendicular to its axis.

Magnetic Moment (M):

$M = \text{pole strength} \times \text{length}$.

Since the new length is L (half of $2L$), the new moment is $m \times L = \frac{M}{2}$.

(NCERT Ed. 2025-26, 12th, Page No. 137)

6. (2)

Series limit corresponds to transition from $n = \infty$.

Frequency $\nu \propto \frac{1}{n^2}$ (where n is the lower orbit).

For Lyman ($n = 1$): $\nu_L \propto \frac{1}{1^2} = 1$ For Pfund ($n = 5$):

$\nu_P \propto \frac{1}{5^2} = \frac{1}{25}$

$\therefore \nu_P = \frac{\nu_L}{25}$

(NCERT Ed. 2025-26, 12th, Page No. 297)

7. (1)

Electric flux $\Phi = \vec{E} \cdot \vec{A} = EA\cos\alpha$, where α is the angle between the field and the area vector.

Since the field is in the plane of the surface, it is perpendicular to the area vector ($\alpha = 90^\circ$).

$\Phi = EA\cos 90^\circ = 0$.

(NCERT Ed. 2025-26, 12th, Page No. 22)

8. (3)

Compare with standard equation:

$E = E_0 \sin(\omega t - kx)$.

$E_0 = 60\text{V/m}$.

$B_0 = \frac{E_0}{c} = 60 / (3 \times 10^8) = 20 \times 10^{-8} = 2 \times 10^{-7}\text{ T}$.

(Incorrect options check: Direction is $+x$ due to minus sign;

$\lambda = 2\pi / k = 2\pi / 500 \approx 1.25\text{cm}$;

$\nu = \omega / 2\pi \approx 23\text{GHz}$.

(NCERT Ed. 2025-26, 12th, Page No. 206)

9. (2)

Since charge at plates in parallel plate capacitor is CV so $i_C = \frac{dq}{dt} = C\frac{dV}{dt} = 6\,\mu\text{A}$

And displacement current is always equal to conduction current.

(NCERT Ed. 2025-26, 12th, Page No. 202)

10. (2)

The magnetic field at point A is zero when the velocity vector of the charge points directly towards or away from A (Biot-Savart Law: $\vec{B} \propto \vec{v} \times \vec{r}$).

From a point A at distance $2R$, two tangents can be drawn to the circle of radius R .

[Contd...

Angle of tangent points relative to center: $\cos\theta = \frac{R}{2R} \Rightarrow \theta = 60^\circ$ (angle with the line connecting centers).

The angle subtended at the center between the two tangent points is $180^\circ - 60^\circ = 120^\circ = \frac{2\pi}{3}$ rad.

$\omega = \frac{\text{Angle}}{t} = \frac{2\pi/3}{t} = \frac{2\pi}{3t}$

(NCERT Ed. 2025-26, 12th, Page No. 113)

11. (4)

According to Gauss's Law, $q_{\text{enclosed}} = \epsilon_0 \Phi$.

At the surface of the sphere ($r = a$), the field $E = Aa$.

Flux $\Phi = E \times \text{Area} = (Aa) \times (4\pi a^2) = 4\pi A a^3$.

$q = \epsilon_0 (4\pi A a^3) = 4\pi \in_0 Aa^3$

(NCERT Ed. 2025-26, 12th, Page No. 30)

12. (2)

For an observer standing at the center (midpoint) between a mirror and a wall, the minimum mirror height (h_m) required to see the full wall is one-third of the wall's height. $h_m = \frac{H_{wall}}{3}$, $h_m = \frac{4.5}{3} = 1.5\text{m}$

(NCERT Ed. 2025-26, 12th, Page No. 222)

13. (2)

For a current $I = I_{DC} + I_0 \sin(\omega t + \phi)$:

$I_{rms} = \sqrt{I_{DC}^2 + \frac{I_0^2}{2}}$

$I_{rms} = \sqrt{3^2 + \frac{4^2}{2}} = \sqrt{9+8} = \sqrt{17}$ A

(NCERT Ed. 2025-26, 12th, Page No. 180)

14. (1)

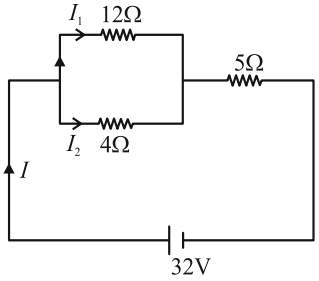
Distance of A from origin: $r_A = \sqrt{3^2 + 4^2} = 5$ m.

Distance of B from origin: $r_B = 5$ m. Since $r_A = r_B$, potential $V = \frac{kQ}{r}$ is the same.

Thus, $V_A - V_B = 0$.

(NCERT Ed. 2025-26, 12th, Page No. 48)

15. (4)



$32 = I R_{eq}$

$R_{eq} = \frac{12 \times 4}{12+4} + 5 = 8\Omega$

So, $I = 4$ A

$I_1 = \frac{4}{4+12} \times I = \frac{4}{16} \times 4 = 1$ A.

(NCERT Ed. 2025-26, 12th, Page No. 95)

16. (4)

For a voltmeter: $V = I_g (G + R_{ext})$

$10 = 2 \times 10^{-3} (G + 4950)$

$5000 = G + 4950 \Rightarrow G = 50\Omega$

(NCERT Ed. 2025-26, 12th, Page No. 131)

17. (2)

3rd line of Balmer: Transition $n = 5 \to n = 2$

Energy of photon $E = \frac{12375}{1085} \approx 11.4$ eV

$E = E_{ground} \left(\frac{1}{2^2} - \frac{1}{5^2} \right) = E_{ground} \left(\frac{21}{100} \right)$

$11.4 = E_{ground} (0.21)$

$E_{ground} = \frac{11.4}{0.21} \approx 54.3$ eV

(NCERT Ed. 2025-26, 12th, Page No. 297)

18. (2)

Gauss's law for magnetism states that isolated magnetic monopoles do not exist. Therefore, the total magnetic flux through any closed surface is always zero ($\oint \vec{B} \cdot d\vec{A} = 0$).

(NCERT Ed. 2025-26, 12th, Page No. 137)

19. (2)

Change in flux $\Delta\phi = \text{Charge}(q) \times \text{Resistance}(R)$

Since current is constant,

$q = i \times t = 10 \text{ mA} \times 5 \text{ s} = 50 \text{ mC}$

$\Delta\phi = (50 \times 10^{-3}) \times 0.5 = 25 \times 10^{-3}$ Wb

$= 25$ mWb

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[Contd...

20. (1)

Using the Law of Mass Action: $n_e n_h = n_i^2$

$$n_e = \frac{n_i^2}{n_h} = \frac{(1.5\times10^{16})^2}{4.5\times10^{22}}$$

$$n_e = \frac{2.25\times10^{32}}{4.5\times10^{22}} = 0.5\times10^{10} = 5\times10^{9}\ \text{m}^{-3}$$

(NCERT Ed. 2025-26, 12th, Page No. 332)

21. (4)

Upright and diminished ($m<1$) image implies a Concave lens forming a virtual image. For virtual images, object and image are on the same side. Distance $=|u|-|v|=20$.

$m=\frac{v}{u}=\frac{1}{2}\Rightarrow u=2v$.

$2v-v=20\Rightarrow v=20$ cm, $u=40$ cm.

Using sign convention (concave lens):

$u=-40, v=-20$.

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}=\frac{1}{-20}-\frac{1}{-40}=\frac{-2+1}{40}=-\frac{1}{40}.$$

$f=-40$ cm

(NCERT Ed. 2025-26, 12th, Page No. 235)

22. (3)

$$C_{air}=\frac{\epsilon_0 A}{d}=4$$

$$C_{new}=\frac{K\epsilon_0 A}{d/2}=2K\left(\frac{\epsilon_0 A}{d}\right)=2K(4)$$

$40=8K\Rightarrow K=5$.

(NCERT Ed. 2025-26, 12th, Page No. 70)

23. (1)

Condition for coincidence: $n_1\lambda_1=n_2\lambda_2$

$$\frac{n_1}{n_2}=\frac{\lambda_2}{\lambda_1}=\frac{5200}{6500}=\frac{4}{5}$$

Least distance corresponds to $n_1=4$

(for $\lambda_1=6500$ Å).

$$y=\frac{n_1\lambda_1 D}{d}=\frac{4\times6500\times10^{-10}\times1.2}{2\times10^{-3}}$$

$y=15.6\times10^{-4}$ m $=1.56$ mm $=0.156$ cm

(NCERT Ed. 2025-26, 12th, Page No. 266)

24. (2)

For unlike charges, the null point is outside the smaller charge. Let distance from $-q$ be x.

$$E_1=E_2\Rightarrow\frac{k(9q)}{(16+x)^2}=\frac{kq}{x^2}$$

$$\frac{3}{16+x}=\frac{1}{x}\Rightarrow 3x=16+x\Rightarrow 2x=16\Rightarrow x=8\text{ cm}$$

(NCERT Ed. 2025-26, 12th, Page No. 19)

25. (3)

Energy of photons: $E_1=\frac{12400}{3100}=4\text{eV}$ and

$E_2=\frac{12400}{6200}=2\text{eV}$.

Since $v_1:v_2=2:1$, the ratio of kinetic energies is

$K_1:K_2=4:1$

Using $K=E-\phi$:

$4-\phi=4(2-\phi)$

$4-\phi=8-4\phi\Rightarrow 3\phi=4\Rightarrow\phi=\frac{4}{3}\text{eV}$

(NCERT Ed. 2025-26, 12th, Page No. 281)

26. (3)

Stability increases when Binding Energy (BE) per nucleon increases. In fusion, lighter nuclei combine to form a heavier, more stable nucleus with a higher BE per nucleon, releasing energy.

(NCERT Ed. 2025-26, 12th, Page No. 315)

27. (2)

Change in flux:

$|\Delta\phi|=|BA\cos(180°)-BA\cos(0°)|=2BA$

$|\Delta\phi|=2\times2\times(10\times10^{-4})=4\times10^{-3}$ Wb

Average EMF $\varepsilon=\frac{|\Delta\phi|}{\Delta t}=\frac{4\times10^{-3}}{0.01}=0.4\text{V}$

(NCERT Ed. 2025-26, 12th, Page No. 171)

28. (2)

Since displacement current will be equal to conduction current so if a variable frequency ac source connected to a capacitor then with decrease in frequency, the displacement current will decrease.

(NCERT Ed. 2025-26, 12th, Page No. 204)

29. (4)

$A\cdot\bar{A}$ is always 0 (since if $A=1,\bar{A}=0$ and vice versa).

All other options evaluate to 1.

30. (3)

Distance between sources $d=\lambda+\lambda=2\lambda$.

Maximum path difference $\Delta x_{max}=d=2\lambda$.

[Contd...

Maxima occur at $\Delta x = 0, \pm\lambda, \pm 2\lambda$.

Points on circle:

$\Delta x = \pm 2\lambda$: 2 points (at ends of diameter passing through sources).

$\Delta x = 0$: 2 points (at ends of diameter perpendicular to sources).

$\Delta x = \pm\lambda$: 4 points (one in each quadrant). Total maxima = $2+2+4=8$.

(NCERT Ed. 2025-26, 12th, Page No. 264)

31. (2)

The first two diodes act as a full-wave rectifier arrangement, ensuring current flows through R_L in the same direction for both half-cycles.

(NCERT Ed. 2025-26, 12th, Page No. 338)

32. (1)

The formula for intensity passing through crossed polarizers with a middle polarizer at angle θ is:

$$I_{final} = \frac{I_0}{8}\sin^2(2\theta)$$

(A) $\theta = 45^\circ$: $I = \frac{I_0}{8}\sin^2(90^\circ) = \frac{I_0}{8}$ (Matches III)

(B) $\theta = 22.5^\circ$: $I = \frac{I_0}{8}\sin^2(45^\circ) = \frac{I_0}{8}\left(\frac{1}{2}\right) = \frac{I_0}{16}$

(Matches I)

(C) $\theta = 30^\circ$: $I = \frac{I_0}{8}\sin^2(60^\circ) = \frac{I_0}{8}\left(\frac{3}{4}\right) = \frac{3I_0}{32}$

(Matches II)

(D) $\theta = 0^\circ$: $I = \frac{I_0}{8}\sin^2(0) = 0$ (Matches IV)

(NCERT Ed. 2025-26, 12th, Page No. 270)

33. (2)

Since the wires are in series, the current (I) is the same. Potential gradient is defined as: $K = \frac{V}{L} = \frac{IR}{L}$

Using $R = \rho\frac{L}{\pi r^2}$, we get: $K = \frac{I\cdot\rho L}{\pi r^2\cdot L} = \frac{I\rho}{\pi r^2}$

For the same material (ρ) and current (I):

$$K \propto \frac{1}{r^2}$$

Given $r_{AB} = 2r_{BC}$: $\frac{K_{AB}}{K_{BC}} = \left(\frac{r_{BC}}{r_{AB}}\right)^2 = \left(\frac{1}{2}\right)^2 = 1:4$

(NCERT Ed. 2025-26, 12th, Page No. 84)

34. (2)

Electric potential is a scalar quantity, so we sum the charges: $Q_{net} = q + 3q = 4q$.

The distance from any point on the ring to the axial point is

$r = \sqrt{R^2 + (3\sqrt{2}R)^2} = \sqrt{R^2 + 18R^2} = \sqrt{19}R$.

Potential $V = \frac{kQ_{net}}{r} = \frac{k(4q)}{\sqrt{19}R}$.

(NCERT Ed. 2025-26, 12th, Page No. 48)

35. (2)

Max Kinetic Energy $K \propto v^2 \propto (h\nu - h\nu_0)$

Case 1: $v_1^2 \propto (10\nu_0 - \nu_0) = 9\nu_0$

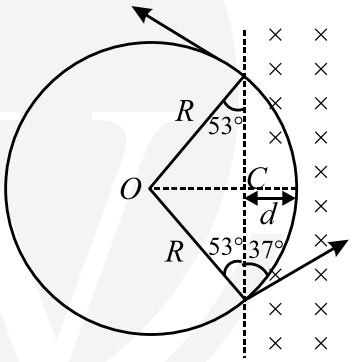
Case 2: $v_2^2 \propto (5\nu_0 - \nu_0) = 4\nu_0$

Ratio: $\frac{v_2}{v_1} = \sqrt{\frac{4}{9}} = \frac{2}{3}$

$v_2 = \frac{2}{3}\times\left(6\times10^6\right) = 4\times10^6$ m/s

(NCERT Ed. 2025-26, 12th, Page No. 281)

36. (1)



$OC = R\sin 53^\circ$

$= R\times\frac{4}{5} = \frac{4R}{5}$

$d = R - \frac{4R}{5} = \frac{R}{5}$

$= \frac{mv}{qB\times 5}$

$= \frac{10^{-6}\times 50}{10^{-6}\times 10\times 5}$

$= 1$ m

(NCERT Ed. 2025-26, 12th, Page No. 112)

37. (4)

Equating energy formulas: $E = mc^2$ and $E = h\nu$

$mc^2 = h\nu$

$\Rightarrow m = \frac{h\nu}{c^2}$

(NCERT Ed. 2025-26, 12th, Page No. 283)

[Contd...

38. (4)

$${}^{A}_{Z}X \xrightarrow{4\alpha,2\beta^-} {}^{206}_{82}pb + 4^4_2He + 2^0_{-1}\beta$$

$A = 206 + 16 = 222$

$Z = 82 + 8 - 2 = 88$

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39. (3)

For a series LCR circuit, the source voltage V is given by: $V^2 = V_R^2 + (V_L - V_C)^2$

Given $V = 50$ V, $V_L = 60$ V, $V_C = 100$ V,

and $R = 10\Omega$: $50^2 = V_R^2 + (60 - 100)^2$

$2500 = V_R^2 + 1600 \Rightarrow V_R^2 = 900 \Rightarrow V_R = 30$ V

Ammeter current $I = \frac{V_R}{R} = \frac{30}{10} = 3$ A

(NCERT Ed. 2025-26, 12th, Page No. 95)

40. (3)

At $t = 0$: Inductor L acts as an open circuit; Capacitor C acts as a short circuit. Current flows only through the capacitor branch. Total resistance $R_{eq} = R + R = 2R$. Thus, $I_0 = \frac{E}{2R}$.

At $t = \infty$: Inductor L acts as a short circuit; Capacitor C acts as an open circuit. Current flows only through the inductor branch. Total resistance $R_{eq} = R + R = 2R$. Thus, $I_\infty = \frac{E}{2R}$.

Ratio: $I_0 : I_\infty = 1:1$.

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41. (1)

Kinetic Energy (KE) depends only on the orbit velocity and is independent of the reference level chosen for Potential Energy.

For $n = 1$ (ground state), $KE = 13.6$ eV.

(NCERT Ed. 2025-26, 12th, Page No. 300)

42. (3)

The equivalent focal length F for two thin lenses in contact is given by: $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$ Since the focal length of a lens depends on its shape and material, not its orientation in contact, and all lenses here are identical ($f_1 = f_2 = f$), the combined focal length in all three cases will be $F = \frac{f}{2}$. The ratio is $\frac{f}{2} : \frac{f}{2} : \frac{f}{2} = 1:1:1$.

(NCERT Ed. 2025-26, 12th, Page No. 238)

43. (1)

Resistance of each part = $\frac{R}{2n}$.

For n such parts connected in series, equivalent resistances, say $R_1 = n \times \left[\frac{R}{2n}\right] = \frac{R}{2}$

Similarly, equivalent resistance, say R_2 , for another set of n identical, respectively, in parallel resistances would be $\frac{1}{n}\left(\frac{R}{2n}\right) = \frac{R}{2n^2}$

For getting maximum of R_1 and R_2 , the resistances should be connected in series and hence,

$$R_{eq} = R_1 + R_2 = \frac{R}{2}\left(1 + \frac{1}{n^2}\right)$$

(NCERT Ed. 2025-26, 12th, Page No. 95)

44. (2)

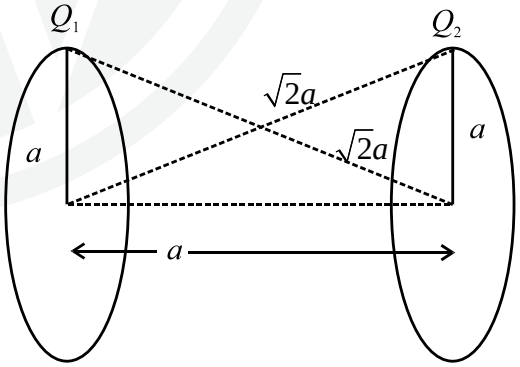
α particles: $\Delta A = 228 - 212 = 16$.

Number of $\alpha = 16/4 = 4$.

β particles: After 4α decays, Z would be $90 - (4 \times 2) = 82$. To reach $Z = 83$, one β^- must be emitted $(83 - 82 = 1)$.

(NCERT Ed. 2025-26, 12th, Page No. 314)

45. (3)



$$V_1 = \frac{1}{4\pi\epsilon_0}\frac{Q_1}{a} + \frac{1}{4\pi\epsilon_0}\frac{Q_2}{a\sqrt{2}}$$

Similarly for 2nd ring (at centre)

$$V_2 = \frac{1}{4\pi\epsilon_0}\frac{Q_1}{a\sqrt{2}} + \frac{1}{4\pi\epsilon_0}\frac{Q_2}{a}$$

Work done $= Q(V_1 - V_2)$

$$= Q\frac{(\sqrt{2}-1)}{4\pi\epsilon_0 a\sqrt{2}}(Q_1 - Q_2)$$

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[Contd...

46. (2)

$T_b - T_f = 105^oC (given)$

$\Delta T_b = K_b \times m$

$T_b - T_b^o = K_b \times m$ .....(1)

$\Delta T_f = K_f \times m$

$T_f^o - T_f = K_f \times m$ .....(2)

Add (1) & (2)

$\left(T_b - T_b^o\right) + \left(T_f^o - T_f\right) = K_b \times m + K_f \times m$

$\left(T_b - T_f\right) - 100 = m\left(K_b + K_f\right)$

$$105 - 100 = \frac{\frac{w}{342}}{\frac{100}{1000}} \times 2.37 \Rightarrow w = 72\,gm$$

(NCERT Ed. 2025-26, 12th, Page No. 18)

47. (3)

For this concentration cell-

$$Cu(s) + \underset{(0.04M)}{Cu^{+2}} \rightarrow \underset{(0.01M)}{Cu^{+2}} + Cu(s)$$

$$E_{cell} = 0 - \frac{RT}{2F}\ln\frac{0.01}{0.04} = -\frac{RT}{2F}\ln(0.5)^2$$

$$E_{cell} = -\frac{RT}{F}\ln(0.5)$$

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48. (3)

$k_1 = Ae^{-Ea_1/RT}; k_2 = Ae^{-E_{a_2}/RT}$

As it is given that activation energy of R_1 then, exceeds R_2 it will be = $E_{a_1} - E_{a_2}$ =10kJ mol^{-1}

The new expression becomes,

$k_1 = Ae^{-Ea_1/RT}; k_2 = Ae^{-\left(E_{a_1} - 10\right)/RT}$

On solving the above equations, we get

$$\ln\left(\frac{k_2}{k_1}\right) = \frac{10}{RT} \Rightarrow \ln\left(\frac{k_2}{k_1}\right) = 4$$

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49. (1)

$N_2O_3 + H_2O \rightarrow 2HNO_2$ (Nitrous acid)

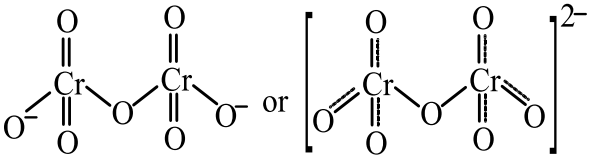
(NCERT Ed. 2022-23, 12th, Page No. 9)

50. (1)

Iodine is solid at room temperature.

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51. (1)



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52. (1)

IUPAC name of $Na_3[CoCl(NO_2)_5]$ is sodium chloridopentanitrocobaltate (III).

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53. (3)

$XeF_6 + 3H_2O \rightarrow XeO_3 + 6HF$

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54. (2)

The atomic and ionic radii of Zr and Hf are equal in size due to lanthanoid contraction.

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55. (2)

Pb_3O_4	Mixed oxide
N_2O	Neutral oxide
Mn_2O_7	Acidic oxide
Bi_2O_3	Basic oxide

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56. (2)

Both the given compound shows ionization isomerism, because they furnish different ion in solution.

$\left[PtCl_2(NH_3)_4\right]Br_2 \rightleftharpoons \left[PtCl_2(NH_3)_4\right]^{2+} + 2Br^-$

$\left[PtBr_2(NH_3)_4\right]Cl_2 \rightleftharpoons \left[PtBr_2(NH_3)_4\right]^{2+} + 2Cl^-$

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57. (2)

The ground state electronic configuration of Th is [Rn] $6d^2\ 7s^2$.

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58. (3)

ionization enthalpy of group 15 element is much greater than that of 14 elements, in corresponding period.

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[Contd...

59. (4)

1 equivalent of any substance is deposited by l F of charge. Calculate the charge on Ca in $CaCl_2$ as:

Cl has –1 charge.

Ca + 2(–l) = 0

Ca = 2

We have to get 20 g Ca from Ca^{2+}.

Number of required moles = mass / molar mass

Molar mass of Ca is 40 g/mol and required mass of Ca is 20 g.

Hence, number of moles = 20/40

= 0.5 mol

Electricity required to produce 1 mol of calcium = 2 F

The electricity required to produce 0.5 mol of calcium = 0.5 × 2 F = 1 F

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60. (3)

Mathematical expression for rate of reaction	Rate law
Rate of reaction for zero order reaction is equal to	Rate constant
Units of rate constant for zero order reaction is same as that of	Rate of reaction
Unit of rate constant for first order reaction	$Second^{-1}$

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61. (2)

Formation of multiple oxoacids by halogens requires the ability to expand the octet and accommodate higher oxidation states through the use of vacant d-orbitals Fluorine belongs to the second period and does not have vacant d-orbital in its valence shell, so it cannot expand its octet. As a result, it forms only one oxoacid (HOF), unlike chlorine, bromine and iodine, which can form several oxoacids due to availability of d-orbitals.

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62. (3)

In acidic medium, CrO_4^{2-} changes to $Cr_2O_7^{2-}$.

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63. (1)

Solution: Although fluorine forms a short bond, the F-F bond is weakened due to strong repulsions between the three lone pairs on each fluorine atom in a very small molecular space. This inter-electronic repulsion reduces the bond strength, making its bond dissociation enthalpy lower than that of Cl-Cl, so both statements are correct.

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64. (4)

Effective atomic number for $[Rh(H_2O)_6]^{3+}$ is calculated as

EAN = z – x + y

where, z = atomic number

x = charge on the metal

y = total electrons donated by ligand

At. No. of Rh = 45

EAN = 45 – 3 + 2 × 6 = 54

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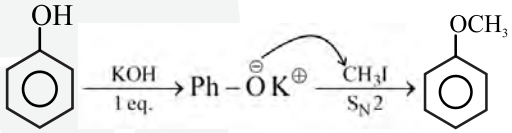
65. (2)

$$Cr(24) \rightarrow [Ar]3d^5 4s^1$$

Cr has higher second ionisation enthalpy as after removing one electron from Cr, the resulting configuration, i.e., [Ar] $3d^5$ becomes more stable.

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66. (3)



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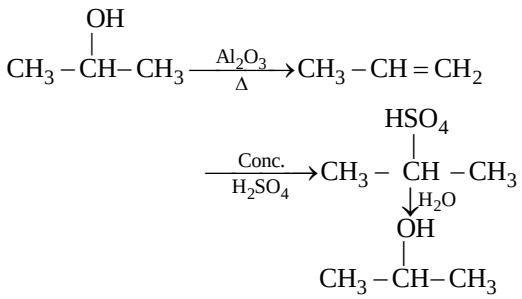
67. (3)

Fe^{2+}: $3d^6 4s^0$

Since, Fe^{2+} has 4 unpaired electrons, hence Fe^{2+} is paramagnetic in nature.

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68. (1)



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69. (3)

The given complex gives white ppt with $AgNO_3$ means one chloride ion is present outside the sphere and the chloride ion is ionizable i.e. free Cl^-

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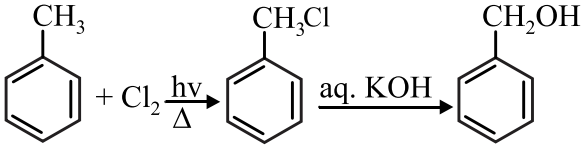
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70. (2)

Benzyl halide are more reactive than vinyl & aryl halide are less reactive than alkyl halide. Hence, option (2) is correct

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71. (4)



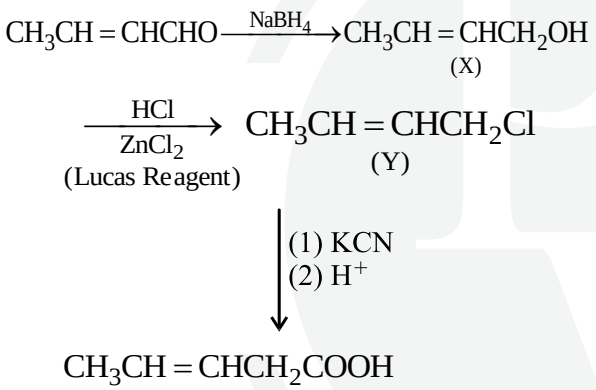
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72. (1)

V_2O_5 and Cr_2O_3 both react with alkalies as well as acids are amphoteric oxides.

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73. (1)



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74. (4)

Higher the number of en(strong ligands), larger will be the splitting.

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75. (1)

Aldehyde & ketone gives 2, 4–DNP test and $\left(CH_3-\underset{\underset{O}{\|}}{C}-\right)$ groups gives positive iodoform test.

Option (1) fulfilling both the conditions.

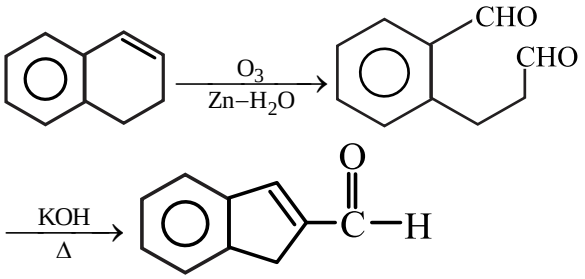
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76. (1)

Aromatic aldehydes and formaldehydes both undergoes cannizaro reaction due to absence of α-H. But formaldehyde is more reactive than benzaldehyde.

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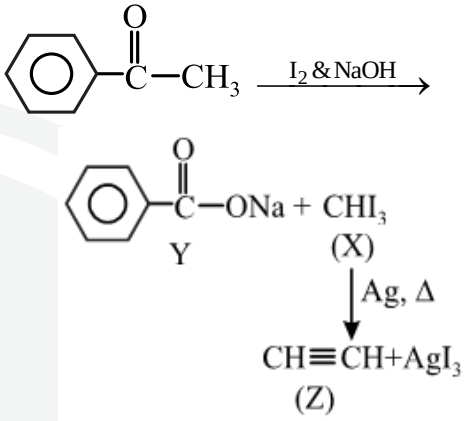
77. (2)



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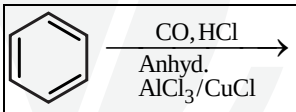
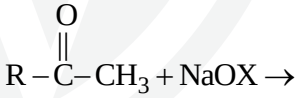
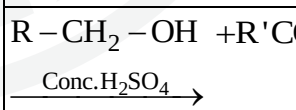
78. (2)

Haloform reaction than formation of acetylene.



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79. (3)

	Gattermann-Koch reaction
	Haloform reaction
	Esterification
	Hell Volhard Zelinsky reaction

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80. (4)

Spectro-chemical series represents the correct order of increasing field strength of ligands to form coordination compounds

$$I^- < Br^- < S^{2-} < SCN^- < Cl^- < N_3^- < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < EDTA^{4-} < NH_3 < en < CN^- < CO$$

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[Contd...

81. (1)

Nb = $[Kr]4d^4 5s^1$, d electrons = 4 = x

Ru = $[Kr]4d^7 5s^1$, d electrons = 7 = y

x + y = 4 + 7 = 11

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82. (2)

Acetaldehyde react with Tollens' reagent to give a silver mirror.

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83. (3)

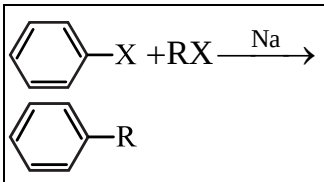
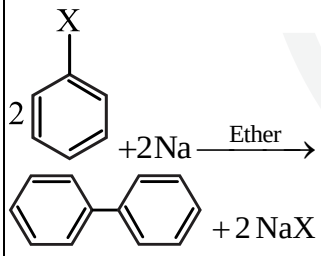
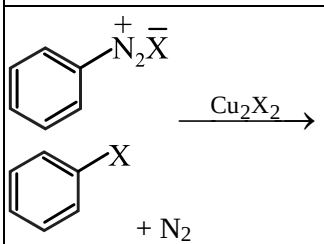
It is Reimer-Tiemann reaction the electrophile formed is :$CC1_2$ (Dichlorocarbene) according to the following reaction

$$CHCl_3 + OH^- \rightleftharpoons \overset{-}{\underset{\cdot\cdot}{C}}Cl_3 + H_2O$$

$$\overset{-}{\underset{\cdot\cdot}{C}}Cl_3 \longrightarrow \underset{\text{Electrophile}}{:CCl_2} + Cl^-$$

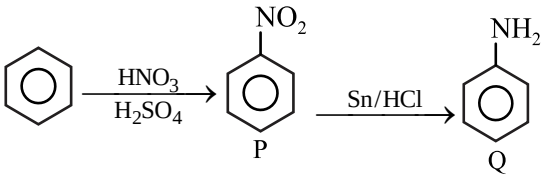
(NCERT Ed. 2025-26, 12th, Page No. 21)

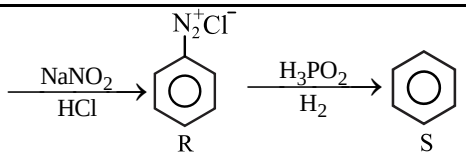
84. (3)

	Wurtz Fitting reaction
	Fitting reaction
	Sandmeyer reaction
$C_2H_5Cl + NaI \xrightarrow{\text{dry acetone}} C_2H_5I + NaCl$	Finkelstein reaction

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85. (2)





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86. (4)

$$\Lambda^o_{m(HAc)} = \Lambda^o_{H^+} + \Lambda^o_{Cl^-} + \lambda^o_{Ac^-} + \lambda^o_{Na^+} - \lambda^o_{Cl^-} - \lambda^o_{Na^+}$$

$$= \Lambda^o_{m(HCl)} + \Lambda^o_{m(NaAc)} - \Lambda^o_{m(NaCl)}$$

$$= (425.9 + 91.0 - 126.4)\,S\ cm^2 mol^{-1}$$

$$= 390.5 S\ cm^2 mol^{-1}$$

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87. (4)

Dimethyl amine is more basic than methylamine.

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88. (4)

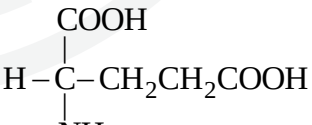
Ethanol-acetone mixture shows positive deviation due to weaker A-B interaction in comparison to A – A or B-B interaction.

Pure ethanol molecules are hydrogen bonded. When acetone is added, its molecules get in between the ethanol molecules and break some of the hydrogen bonds between them.

Thus, the intermolecular attractive interactions are weakens and the solution shows positive deviation from Raoult's law.

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89. (4)



(Glutamic acid)

It is acidic in nature because in this group, there are two carboxylic acid and one amino acid.

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90. (3)

$$k = \frac{2.303}{t}\log\frac{R_o}{R}$$

$$t = \frac{2.303}{6.909}\log\frac{100}{25} = \frac{1}{3}\log 4$$

$$t = \frac{1}{3}\log 2^2 \Rightarrow \frac{2}{3}\log 2$$

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[Contd...

91. (2)

In some cereals such as rice and wheat, pollen grains lose viability within 30 minutes of their release, and in some members of Rosaceae, Leguminoseae and Solanaceae, they maintain viability for months.

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92. (2)

Birth rate is calculated as:

$$\text{Birth rate} = \frac{\text{Number of new individuals}}{\text{Initial population}}$$

Given:

Initial population = 50 lotus plants

New plants added = 10

$$\text{Birth rate} = \frac{10}{50} = 0.2 \text{ offspring per lotus per year}$$

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93. (3)

A bacteriophage known as $\phi \times 174$ has 5386 nucleotides, Bacteriophage lambda has 48502 base pairs (bp), *Escherichia coli* has 4.6×10^6 bp, and haploid content of human DNA is 3.3×10^9 bp.

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94. (3)

Depending on the type of the raw material used for fermentation and the type of processing (with or without distillation) different types of alcoholic drinks are obtained. Wine and beer are produced without distillation whereas whisky, brandy and rum are produced by distillation of the fermented broth.

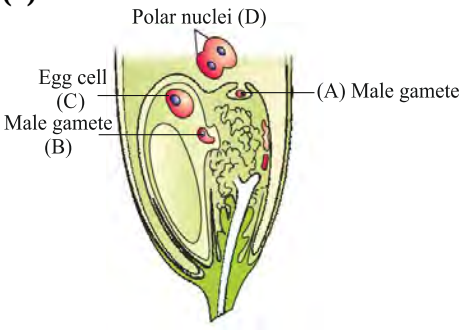
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95. (2)

Vertical distribution of different species occupying different levels is called stratification. For example, trees occupy top vertical strata or layer of a forest, shrubs the second and herbs and grasses occupy the bottom layers in forest.

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96. (4)



Fusion of one male gamete (A) and two polar nuclei (D) results in the formation of primary endosperm nucleus (PEN).

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97. (2)

Har Gobind Khorana	Synthesised RNA molecules with defined combinations of bases
Marshall Nirenberg	Cell-free system for protein synthesis
Severo Ochoa	Used polynucleotide phosphorylase enzyme
Friedrich Miescher	Identified DNA as nuclein

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98. (3)

Tropics shows greater biological diversity than temperate because:

(a) Speciation is generally a function of time, unlike temperate regions subjected to frequent glaciations in the past, tropical latitudes have remained relatively undisturbed for millions of years and thus, had a long evolutionary time for species diversification.

(b) Tropical environments, unlike temperate ones, are less seasonal, relatively more constant and predictable. Such constant environments promote niche specialisation and lead to a greater species diversity.

(c) There is more solar energy available in the tropics, which contributes to higher productivity; this in turn might contribute indirectly to greater diversity.

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99. (1)

Vegetative cell of pollen grain: It is bigger, has abundant food reserve and a large irregularly shaped nucleus.

Generative cell of pollen grain: It is smaller and floats in the cytoplasm of the vegetative cell. It is spindle shaped with dense cytoplasm and a nucleus.

Microspore Mother Cell (MMC): As the anther develops, the cells of the sporogenous tissue undergo meiotic divisions to form microspore tetrads. Each cell of the sporogenous tissue is a potential microspore mother cell.

Central cell of embryo sac: This is the largest cell of the embryo sac. It contains two polar nuclei and, after triple fusion with a male gamete, it becomes the Primary Endosperm Cell (PEC).

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[Contd...

100. (2)

Start (Generation 0):

Bacteria are grown in a $^{15}NH_4Cl$ medium. All DNA molecules are heavy:

$^{15}N–^{15}N$

After 20 Minutes (Generation 1):

The bacteria are transferred to a $^{14}NH_4Cl$ medium. After one division, each DNA molecule consists of one old heavy strand and one new light strand:

$^{15}N–^{14}N$

Thus, DNA is 100% hybrid.

After 40 Minutes (Generation 2):

The two hybrid ($^{15}N–^{14}N$) molecules replicate:

The strands containing ^{15}N form new hybrid molecules:

$^{15}N–^{14}N$

The strands containing ^{14}N form new light molecules:

$^{14}N–^{14}N$

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101. (2)

Rhizobium: These are symbiotic bacteria found in the root nodules of leguminous plants that fix atmospheric nitrogen into organic forms.

Azospirillum: These are free-living bacteria in the soil that can fix atmospheric nitrogen, thereby enriching the nitrogen content of the soil.

Anabaena: *Anabaena* is a cyanobacterium (blue-green alga) that can fix nitrogen. In paddy fields, cyanobacteria serve as an important biofertiliser.

Trichoderma species act as effective biocontrol agents against several plant pathogens, rather than as biofertilisers.

Baculoviruses are used as biocontrol agents in integrated pest management (IPM) programs.

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102. (2)

When genes are located on the same chromosome, some genes are very tightly linked, showing very low recombination, whereas others are loosely linked, showing higher recombination.

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103. (2)

The annual net primary productivity of the whole biosphere is approximately 170 billion tons (dry weight) of organic matter. Of this, despite occupying about 70 per cent of the surface, the productivity of the oceans are only 55 billion tons. Rest of course, is on land.

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104. (2)

One of the problems of hybrids is that hybrid seeds have to be produced every year. If the seeds collected from hybrids are sown, the plants in the progeny will segregate and do not maintain hybrid characters. Production of hybrid seeds is costly and hence the cost of hybrid seeds become too expensive for the farmers. If these hybrids are made into apomicts, there is no segregation of characters in the hybrid progeny.

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105. (3)

In eukaryotes, there are at least three RNA polymerases in the nucleus (in addition to the RNA polymerase found in the organelles). There is a clear cut division of labour. The RNA polymerase I transcribes rRNAs (28S, 18S, and 5.8S), whereas the RNA polymerase III is responsible for transcription of tRNA, 5srRNA, and snRNAs (small nuclear RNAs). The RNA polymerase II transcribes a precursor of mRNA, the heterogeneous nuclear RNA (hnRNA).

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106. (3)

The population is said to be stable when the number of pre-reproductive individuals (the youngest age group) is almost equal to the number of reproductive individuals.

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107. (1)

The sex determination in honey bee is based on the number of sets of chromosomes an individual receives. An offspring formed from the union of a sperm and an egg develops as a female (queen or worker), and an unfertilised egg develops as a male (drone) by means of parthenogenesis. This means that the males have half the number of chromosomes than that of a female. The females are diploid having 32 chromosomes and males are haploid, i.e., having 16 chromosomes. This is called as haplodiploid sex-determination system and has special characteristic features such as the males produce sperms by mitosis.

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[Contd...

108. (1)

The primary effluent is passed into large aeration tanks where it is constantly agitated mechanically and air is pumped into it. This allows vigorous growth of useful aerobic microbes into flocs (masses of bacteria associated with fungal filaments to form mesh like structures).

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109. (3)

Some examples of recent extinctions include the dodo (Mauritius), quagga (Africa), thylacine (Australia), Steller's Sea Cow (Russia) and three subspecies (Bali, Javan, Caspian) of tiger.

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110. (3)

The pitch of the helix is 3.4 nm (a nanometre is one billionth of a metre, that is 10^{-9} m) and there are roughly 10 bp in each turn.

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111. (4)

Producing unisexual flowers: If male and female flowers are on the same plant (monoecy, like castor and maize), it prevents autogamy but not geitonogamy.

Self-incompatibility: This is a genetic mechanism that prevents self-pollen (from the same flower or other flowers of the same plant) from fertilizing the ovules by inhibiting pollen germination or pollen tube growth in the pistil.

Producing bisexual cleistogamous flowers: Cleistogamous flowers never open; therefore, they *ensure* self-pollination (autogamy) and are an inbreeding device, not an outbreeding one.

Synchronization of pollen release and stigma receptivity: Synchronization promotes self-pollination. For outbreeding, plants develop dichogamy, where pollen release and stigma receptivity are not synchronized.

Dioecy: This is a condition where male and female flowers are present on different plants (e.g., papaya). This prevents both autogamy and geitonogamy.

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112. (2)

Human genome sequencing involved two major approaches. One approach focused on identifying all the genes that are expressed as RNA (referred to as Expressed Sequence Tags (ESTs). The other took the blind approach of simply sequencing the whole set of genome that contained all the coding and non-coding sequence, and later assigning different regions in the sequence with functions (a term referred to as Sequence Annotation).

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113. (3)

The BOD test measures the rate of uptake of oxygen by micro-organisms in a sample of water and thus, indirectly, BOD is a measure of the organic matter present in the water. The greater the BOD of waste water, more is its polluting potential.

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114. (2)

In a particular climatic condition, decomposition rate is slower if detritus is rich in lignin and chitin, and quicker, if detritus is rich in nitrogen and water-soluble substances like sugars.

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115. (2)

In 1900, three Scientists (de Vries, Correns and von Tschermak) independently rediscovered Mendel's results on the inheritance of characters.

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116. (1)

A true breeding line is one that, having undergone continuous self-pollination, shows the stable trait inheritance and expression for several generations.

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117. (4)

In *lac* operon (here lac refers to lactose), a polycistronic structural gene is regulated by a common promoter and regulatory genes. The *z* gene codes for beta-galactosidase (β-gal), which is primarily responsible for the hydrolysis of the disaccharide, lactose into its monomeric units, galactose and glucose. The *y* gene codes for permease, which increases permeability of the cell to β-galactosides. The *a* gene encodes a transacetylase.

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118. (2)

In nature, a given habitat has limited resources (such as food and space) to support a certain number of individuals of a population beyond which no further growth is possible. This limit is called the carrying capacity (K) for that species in that habitat.

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119. (4)

Wind pollination requires that the pollen grains are light and non-sticky so that they can be transported in wind currents. They often possess well-exposed stamens (so that the pollens are easily dispersed into wind currents) and large often-feathery stigma to easily trap air-borne pollen grains. Wind pollinated flowers often have a single ovule in each ovary and numerous flowers packed into an inflorescence. Both wind and water pollinated flowers are not very colourful and do not produce nectar.

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120. (3)

Chromosome 1 has most genes (2968), and the Y has the fewest (231).

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121. (2)

A small part of the activated sludge is pumped back into the aeration tank to serve as the inoculum. The remaining major part of the sludge is pumped into large tanks called anaerobic sludge digesters. Here, other kinds of bacteria, which grow anaerobically, digest the bacteria and the fungi in the sludge. During this digestion, bacteria produce a mixture of gases such as methane, hydrogen sulphide and carbon dioxide. These gases form biogas and can be used as source of energy as it is inflammable.

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122. (4)

Except for the deep sea hydro-thermal ecosystem, sun is the only source of energy for all ecosystems on Earth. All organisms are dependent for their food on producers, either directly or indirectly. In nature there is an unidirectional flow of energy from the sun to producers and then to consumers. Further, ecosystems are not exempt from the Second Law of thermodynamics. They need a constant supply of energy to synthesise the molecules they require, to counteract the universal tendency toward increasing disorderliness.

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123. (2)

Histones are rich in the basic amino acid residues lysine and arginine.

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124. (3)

Incomplete dominance	Heterozygous condition shows intermediate expression not resembling either parent
Linkage	Physical association of genes
Pleiotropy	Single gene affecting multiple phenotypes
Polygenic inheritance	Multiple genes affecting a single phenotype

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125. (1)

In bacteria, since the mRNA does not require any processing to become active, and also since transcription and translation take place in the same compartment (there is no separation of cytosol and nucleus in bacteria), many times the translation can begin much before the mRNA is fully transcribed. Consequently, the transcription and translation can be coupled in bacteria.

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126. (4)

The large holes in 'Swiss cheese' are due to production of a large amount of CO_2 by a bacterium named *Propionibacterium sharmanii*. The 'Roquefort cheese' are ripened by growing a specific fungi on them, which gives them a particular flavour.

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127. (2)

Biologists are not sure about how many prokaryotic species there might be. The problem is that conventional taxonomic methods are not suitable for identifying microbial species and many species are simply not culturable under laboratory conditions. If we accept biochemical or molecular criteria for delineating species for this group, then their diversity alone might run into millions.

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128. (2)

Klinefelter's Syndrome is caused due to the presence of an additional copy of X chromosome resulting into a karyotype of 47, XXY. Such an individual has overall masculine development, however, the feminine development (development of breast, i.e., Gynaecomastia) is also expressed. Such individuals are sterile.

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129. (2)

During transcription, the mRNA sequence is identical to the coding (sense/non-template) strand, except that Uracil (U) replaces Thymine (T).

Template strand (3′ → 5′): Read by RNA polymerase.

Coding strand (5′ → 3′): Has the same sequence as mRNA (except T → U).

To obtain mRNA from the coding strand, replace T with U. The direction (5′ → 3′) remains the same.

Example:

Coding strand: 5′-T A C G T A C-3′

mRNA: 5′-U A C G U A C-3′

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130. (4)

An affected father and an unaffected mother produce children who do not show the trait. This apparent "skipping" of a generation is a characteristic of recessive inheritance.

All daughters in Generation II are carriers because they inherit the affected chromosome from their father.

The trait reappears in the grandsons. Carrier mothers pass the affected chromosome to approximately 50% of their sons. Since males possess only one X chromosome, they express the trait if they inherit the affected allele.

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131. (4)

The female mosquito is not considered a parasite, although it needs our blood for reproduction. The human liver fluke (a trematode parasite) depends on two intermediate hosts (a snail and a fish) to complete its life cycle.

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132. (4)

Pollen grain has a prominent two-layered wall. The hard outer layer called the exine is made up of sporopollenin which is one of the most resistant organic material known. The inner wall of the pollen grain is called the intine. It is a thin and continuous layer made up of cellulose and pectin.

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133. (3)

When the small subunit of ribosome encounters an mRNA, the process of translation of the mRNA to protein begins.

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134. (2)

A molecule that can act as a genetic material must fulfill the following criteria: (i) It should be able to generate its replica (Replication). (ii) It should be stable chemically and structurally. (iii) It should provide the scope for slow changes (mutation) that are required for evolution. (iv) It should be able to express itself in the form of 'Mendelian Characters'.

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135. (3)

The salient features of genetic code are as follows: (i) The codon is triplet. 61 codons code for amino acids and 3 codons do not code for any amino acids, hence they function as stop codons. (ii) Some amino acids are coded by more than one codon, hence the code is degenerate. (iii) The codon is read in mRNA in a contiguous fashion. There are no punctuations. (iv) The code is nearly universal: for example, from bacteria to human UUU would code for Phenylalanine (phe). Some exceptions to this rule have been found in mitochondrial codons, and in some protozoans. (v) AUG has dual functions. It codes for Methionine (met), and it also act as initiator codon. (vi) UAA, UAG, UGA are stop terminator codons.

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136. (3)

In agarose gel electrophoresis, DNA fragments separate based on size because DNA is negatively charged and moves towards the anode. After running the gel, bands are not directly visible. Staining with ethidium bromide and viewing under UV light shows bright orange DNA bands.

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137. (2)

List-I	List-II
Wombat	Australian marsupial
Lemur	Placental mammal
Microsporum	Fungal pathogen
Plasmodium	Protozoan pathogen

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138. (1)

India started one of the earliest family planning programmes in 1951. This was a major step to control population growth by promoting planned parenthood.

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139. (1)

First, simple organic monomers (like amino acids) formed on early Earth. Then these monomers joined to form polymers (complex molecules). Next, non-cellular life-like giant molecules (RNA/proteins) appeared, and finally the first cellular life originated.

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140. (2)

The first human hormone produced using recombinant DNA technology was insulin. The human insulin gene was introduced into microbes so they could synthesize insulin in large quantities for medical use.

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141. (4)

The primary oocyte completes meiosis-I just before ovulation (not within the secondary follicle as stated). Meiosis-I in oogenesis is unequal, producing a secondary oocyte and a polar body.

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142. (2)

Bt toxin is produced by the bacterium *Bacillus thuringiensis*. Its genes are used in Bt crops to kill insect larvae by damaging their gut after ingestion.

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143. (1)

The shown flowering branch is of opium poppy (*Papaver somniferum*), from which morphine is obtained. Morphine is an analgesic drug derived from the latex of the poppy plant.

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144. (1)

Rosie, the first transgenic cow, produced milk enriched with human alpha-lactalbumin. This showed how transgenic animals can be used to produce human proteins in milk (biopharming).

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145. (1)

Vaccination works because the immune system develops memory after exposure to antigen. Vaccines stimulate memory B and T cells, so on later exposure the body responds faster and stronger, preventing disease.

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146. (4)

Foetal ejection reflex is the signal from the fully developed foetus/placenta that triggers strong uterine contractions. It stimulates release of oxytocin from the maternal posterior pituitary, which increases contractions and helps in child birth.

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147. (1)

Untreated PID can damage the oviducts, causing blockage or improper transport of the zygote. This can lead to ectopic pregnancy, where implantation occurs outside the uterus, commonly in the fallopian tube.

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148. (4)

List-I (PCR Step/ Component)	List-II (Feature/Function)
Denaturation	Separation of the two strands of DNA by high heat
Annealing	Hybridisation of two sets of primers
Extension	Synthesis of DNA using nucleotides at 72°C
Thermus aquaticus	Isolation source of thermostable DNA polymerase

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149. (1)

Sperm move from seminiferous tubules into rete testis, then through vasa efferentia to the epididymis for maturation, and finally enter vas deferens for transport.

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150. (3)

The "good humour" idea was rejected after William Harvey discovered blood circulation. His work showed blood moves in a closed circulatory system rather than being controlled by body "humours."

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151. (4)

Transformation means introducing foreign DNA into a host cell (like bacteria) so the host can express or replicate that DNA. This is a key step in recombinant DNA technology and genetic engineering.

Elution refers to the extraction of DNA bands from the agarose gel (cutting out the piece of gel and extracting DNA). The process of cutting DNA into fragments is done by Restriction Endonucleases.

Downstream processing refers to the separation and purification steps that happen after the biosynthetic stage is complete. The biosynthetic stage itself occurs during fermentation in the bioreactor.

Bioreactors are vessels used for processing large volumes (100–1000 liters) of culture, not small volumes.

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152. (1)

List-I	List-II
Non-medicated IUD	Lippes loop
Hormone releasing IUD	LNG-20
Copper releasing IUD	Multiload 375
Oral Pill	Saheli

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153. (4)

The period between the Devonian and Permian is the Carboniferous. It is known for extensive forests and major coal formation, and it lies chronologically after Devonian and before Permian.

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154. (4)

The Java fossil found in 1891 with about 900 cc cranial capacity is *Homo erectus*. It is an extinct hominid known for more advanced posture and tool use than earlier forms.

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155. (2)

Morula is an early embryonic stage with about 8 to 16 blastomeres formed after cleavage. It later develops into blastocyst as cells differentiate into trophoblast and inner cell mass.

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156. (1)

A grafted kidney is mainly rejected due to cell-mediated immunity. T-lymphocytes recognize foreign antigens on the graft and attack the transplanted tissue, causing rejection.

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157. (2)

Amniocentesis is legally banned in India for sex determination, adoption is legally allowed and is an accepted method for couples who want parenthood.

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158. (3)

Agrobacterium tumefaciens is mainly used as a gene delivery system for plant cells, not for both prokaryotic and eukaryotic hosts broadly. It transfers T-DNA from its Ti plasmid into plant cells, and modified Ti plasmid is made non-pathogenic for gene transfer.

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159. (4)

Hardy–Weinberg equilibrium is disturbed by gene migration (gene flow), genetic drift, mutation, and recombination. Random mating is actually a condition that helps maintain equilibrium, not a disturbing factor.

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160. (1)

List-I	List-II
Gene therapy	Replacement of a faulty gene by a normal healthy gene
Gene cloning	To obtain identical copies of a particular DNA molecule
ELISA	Antigen-antibody interaction
Genetically engineered Human Insulin	*E. coli*

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161. (3)

In a regular 28-day cycle, ovulation occurs around day 14 and is triggered by the LH surge. Therefore, the LH surge most likely induces rupture of the Graafian follicle around Day 14.

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162. (4)

MTP (medical termination of pregnancy) is considered relatively safe during the first trimester when performed by qualified medical professionals.

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163. (4)

List-I	List-II
Elephantiasis	*Wuchereria bancrofti*
Pneumonia	*Haemophilus influenzae*
Cancer	Radiotherapy
AIDS	Viral disease

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164. (2)

Bioreactors require foam control because foaming can interfere with aeration and contamination control. So, the statement saying bioreactors have agitator and oxygen delivery but no foam control system is incorrect.

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165. (4)

Sporozoites are the infective stage for humans, injected by female *Anopheles*. When infected RBCs rupture, they release haemozoin, which causes fever and chills.

Malignant malaria is caused by *Plasmodium falciparum*. The *Plasmodium* reproduces asexually in the liver cells. The female *Anopheles* mosquito is the vector and primary host for malaria.

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166. (2)

List-I	List-II
Thorns of *Bougainvillea* and tendrils of *Cucurbita*	Homologous organs
Eyes of octopus and mammals	Analogous organs
Genetic drift	Change in gene frequency by chance
Gene flow	Transfer of genes into or out of population

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167. (3)

Label X indicates the middle piece of sperm which contains many mitochondria. These mitochondria produce ATP to provide energy for tail movement and help sperm motility.

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168. (4)

Heroin is a depressant drug that slows body functions by acting on the nervous system. It reduces alertness and can cause drowsiness and respiratory depression, so this statement is correct.

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[Contd...

169. (2)

Traditional hybridization often leads to the inclusion and multiplication of undesirable genes along with the desired ones. Globally, about 30 recombinant therapeutics have been approved for human use. In India, 12 of these are currently being marketed. Recombinant DNA technology is specifically designed to introduce only desirable genes into the target organism, avoiding the pitfalls of traditional breeding. Most restriction enzymes are isolated from bacteria, where they serve as a defense mechanism against bacteriophages (viruses).

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170. (1)

The universe is estimated to be about 13.8 billion years old. Other options are incorrect because Earth formed about 4.5 billion years ago (not million), and Pasteur disproved spontaneous generation.

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171. (2)

Spermiogenesis is the transformation of spermatids into spermatozoa, not their release. Release of sperm from Sertoli cells into seminiferous tubules is called spermiation.

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172. (3)

A foreign DNA fragment usually cannot multiply if it stays in cytoplasm without proper replication origin. If it integrates into host chromosome and uses the host origin of replication, it can replicate and be inherited.

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173. (3)

The idea "Ontogeny repeats Phylogeny" was rejected by Karl Ernst von Baer. He stated that embryos of higher animals do not pass through adult stages of lower animals, only show early similarities.

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174. (4)

Transgenic plants are produced by introducing foreign DNA into a plant cell and then regenerating a complete plant from that transformed cell. This is the basis of genetic engineering in plants.

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175. (4)

DNA fragments produced by restriction enzymes are separated by electrophoresis. Because DNA is negatively charged, it moves through a gel towards the positive electrode and separates mainly by size.

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176. (4)

In IUT (intrauterine transfer), an embryo with more than 8 blastomeres is transferred into the uterus for implantation. ZIFT is used when the embryo has up to 8 blastomeres and is placed in fallopian tube.

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177. (1)

List-I	List-II
Labia majora	Fleshy folds of tissue, which extend down from the mons pubis and surround the vaginal opening
Labia minora	Paired folds of tissue under the labia majora
Mons pubis	Cushion of fatty tissue covered by skin and pubic hair
Hymen	A membrane that often partially covers the opening of the vagina

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178. (2)

HIV mainly infects helper T (T_H) cells (also macrophages) and replicates inside them. It gradually destroys these cells, weakening immunity and leading to AIDS.

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179. (3)

Amniocentesis is used to detect chromosomal disorders and genetic diseases like Down syndrome, haemophilia and sickle-cell anaemia. Jaundice is not diagnosed by amniocentesis as a standard genetic/foetal chromosomal test.

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180. (3)

Sticky ends form when some restriction enzymes cut DNA in a staggered manner. The reason given is false because not all restriction enzymes cut away from the centre to create sticky ends; some cut straight and form blunt ends.

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